AI Room Survey - Prototype Overview

We have a working prototype. A customer records a short video of a room on their phone, and the AI lists the furniture with sizes and the total shipment volume.

Where it stands today

- It works from start to finish, and anyone can test it on their phone through a link.
- It runs on my laptop for now, so it is slow: about 2 minutes per room. On a cloud server it should take under a minute.
- Sizes look realistic on what we have tested so far. Current estimate: a room's total is within about 20-25%.
- The customer can fix any size, remove items, or add missed ones before getting a quote.

Our approach

- The customer uses their phone's browser. No app to download, and it works on any iPhone or Android.
- The AI runs on a server, so even old phones work.
- We use free open-source AI, not paid AI services, so there are no per-use AI fees.

How it works

1. The customer records a 20-60 second video of the room, or uploads one.
2. The server takes about 5 frames per second from the video.
3. **Find the items:** YOLOE-26 detects and outlines each piece of furniture from a list of about 40 household items (sofa, bed, wardrobe, table and so on). A tracker (BoT-SORT) follows each item from frame to frame so the same sofa is counted once.
4. **Measure depth:** for the clearest views of each item, MoGe-2 (by Microsoft) estimates the real distance of every pixel in metres. This turns a normal photo into a 3D point cloud, with no special camera or LiDAR needed.
5. **Get the size:** we find the floor, cut out each item's 3D points, and fit a box around them to get length, width and height. Results from several views are combined.
6. **Check the result:** if a door is visible, its standard height (2.03 m) is used to correct the scale. Every size is checked against normal ranges for that item type.
7. The website shows the item list with sizes, volume in cubic feet, and the room total.

Tech stack

- **AI models:** YOLOE-26 (Ultralytics) for detection and MoGe-2 (Microsoft) for depth. Both are open-source and run on our own server.
- **AI server:** Python, PyTorch, FastAPI, OpenCV, FFmpeg.

- **Website:** Next.js (React, TypeScript, Tailwind CSS), hosted on Vercel.
- **GPU hosting (planned):** Modal, a pay-per-second cloud GPU (Nvidia T4 or L4). We only pay while a video is being processed.
- **Right now:** my laptop's GPU (GTX 1650), shared through a Cloudflare link for testing.

What it will cost

- Testing now: $0.
- Cloud AI: about 1-2 cents per room. The cloud provider gives $30 of free usage every month, which covers roughly 1,500 rooms (about 300 homes).
- Website hosting: $20 a month.
- AI token fees: none.
- Licence: YOLOE-26 needs a paid Ultralytics licence for commercial use, or about 1 week of work to switch to a free alternative.